

B.Tech. Degree V Semester First Internal Examination October 2024

Department of Mechanical Engineering (B Batch)

19-200-0501 NUMERICAL AND STATISTICAL METHODS

Time: 2 Hours

Maximum Marks: 40

CO1: Solve algebraic and transcendental equations by numerical methods.

CO2: Solve numerical differentiation and integration problems.

PART A

(4x4=16)

I		Marks	CO	BL																
a ✓	Find a positive value of $(18)^{1/3}$ correct to three places of decimal using Newton-Raphson method.	4	CO1	L2																
b ✓	Find the third divided difference of $f(x)$ with arguments 0,1,2,4 where $f(x) = x^3 - 9x^2 + 21x + 1$.	4	CO1	L2																
c ✓	Use Euler's Method to find $y(0.2)$ given $y' = x^2 + y$, $y(0) = 1$ (Take $h = 0.1$)	4	CO2	L2																
d ✓	The table below gives the velocity v of a moving particle at time t seconds. Find the distance covered by the particle in 12 seconds. <table><tr><td>t:</td><td>0</td><td>2</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td></tr><tr><td>v:</td><td>4</td><td>6</td><td>16</td><td>34</td><td>60</td><td>94</td><td>136</td></tr></table>	t:	0	2	4	6	8	10	12	v:	4	6	16	34	60	94	136	4	CO2	L3
t:	0	2	4	6	8	10	12													
v:	4	6	16	34	60	94	136													

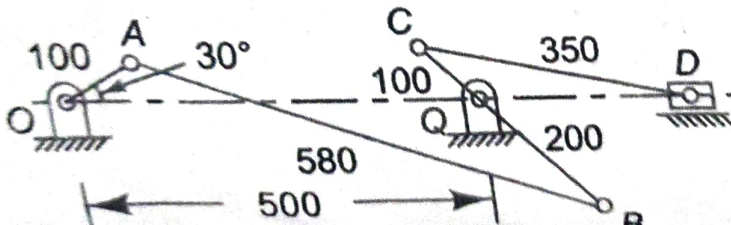
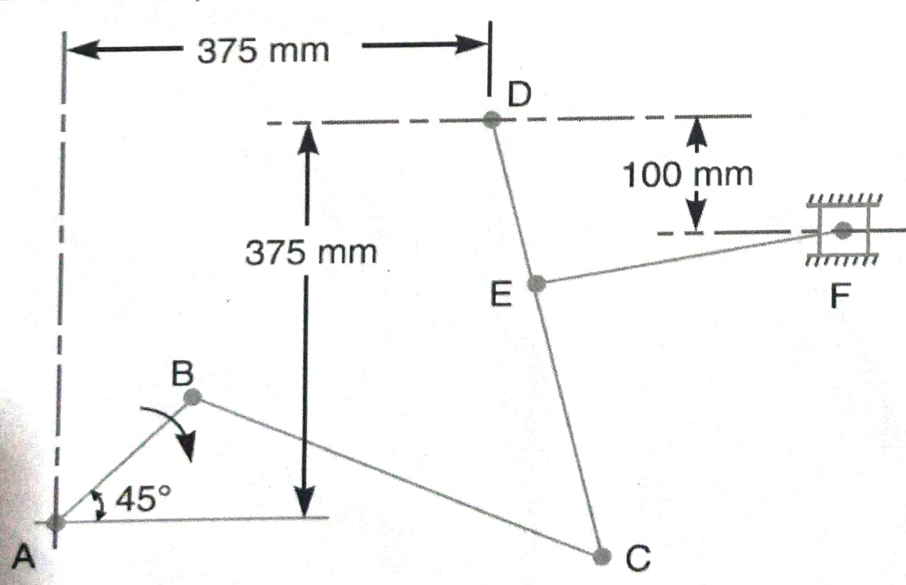
PART B

(2x12=24)

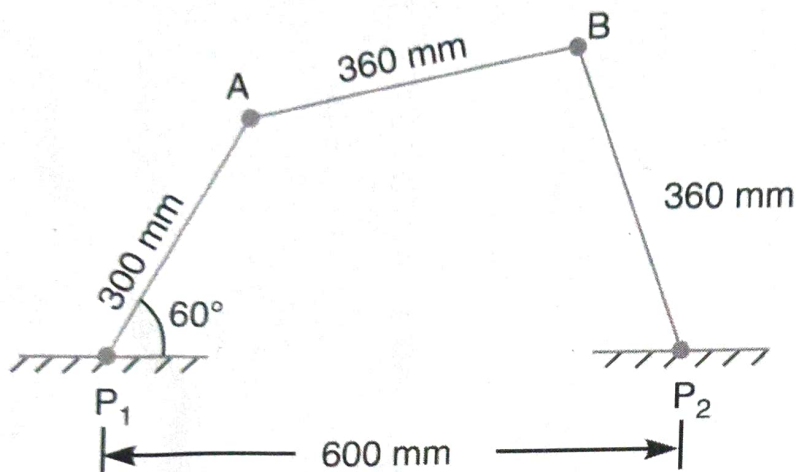
II a	Find a real root of the equation $x^3 - 4x - 9 = 0$ using Regula-Falsi method correct to 4 decimal places.	6	CO1	L2																
b	Solve the system of equation by Gauss Seidel method: $8x + y + z = 8$, $2x + 4y + z = 4$, $x + 3y + 5z = 5$.	6	CO1	L2																
OR																				
IIIa	Find $y(3)$ and $y(22)$ from the following table <table style="margin: 10px auto; border-collapse: collapse;"> <tr> <td>x :</td> <td>0</td> <td>5</td> <td>10</td> <td>15</td> <td>20</td> <td>25</td> </tr> <tr> <td>y :</td> <td>7</td> <td>11</td> <td>14</td> <td>18</td> <td>24</td> <td>32</td> </tr> </table>	x :	0	5	10	15	20	25	y :	7	11	14	18	24	32	6	CO1	L2		
x :	0	5	10	15	20	25														
y :	7	11	14	18	24	32														
b	Find $y(27)$ given <table style="margin: 10px auto; border-collapse: collapse;"> <tr> <td>x:</td> <td>14</td> <td>17</td> <td>31</td> <td>35</td> </tr> <tr> <td>f(x):</td> <td>68.7</td> <td>64</td> <td>44</td> <td>39.1</td> </tr> </table>	x:	14	17	31	35	f(x):	68.7	64	44	39.1	6	CO1	L2						
x:	14	17	31	35																
f(x):	68.7	64	44	39.1																
IVa	A slider in a machine moves along a fixed straight rod. Its distance x cm along the rod is given below for various values of the time ' t ' seconds. Find the velocity of the slider and its acceleration when $t = 0.3$ seconds. <table style="margin: 10px auto; border-collapse: collapse;"> <tr> <td>t :</td> <td>0</td> <td>0.1</td> <td>0.2</td> <td>0.3</td> <td>0.4</td> <td>0.5</td> <td>0.6</td> </tr> <tr> <td>x :</td> <td>30.13</td> <td>31.62</td> <td>32.87</td> <td>33.64</td> <td>33.95</td> <td>33.81</td> <td>33.24</td> </tr> </table>	t :	0	0.1	0.2	0.3	0.4	0.5	0.6	x :	30.13	31.62	32.87	33.64	33.95	33.81	33.24	6	CO2	L3
t :	0	0.1	0.2	0.3	0.4	0.5	0.6													
x :	30.13	31.62	32.87	33.64	33.95	33.81	33.24													
b	Using Taylor's series method, find y at $x = 0.1$ given $\frac{dy}{dx} - 2y = 3e^x$, $y(0) = 0$.	6	CO2	L2																
OR																				

Va	Using Modified Euler's method, find $y(0.2)$ given $\frac{dy}{dx} = y - x^2$, $y(0) = 1$	6	CO2	L2
b	Find $y(0.2)$ given $\frac{dy}{dx} = y - x$, $y(0) = 2$ using R- K method of fourth order.	6	CO2	L2

Time: 2hrs	Department of Mechanical Engineering, SOE First Internal Examination of S-5 B.Tech Mechanical Engineering – October 2024 19-205-0502 MECHANICS OF MACHINERY (B section)	Max Marks:40
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Q No	ANSWER ANY FOUR QUESTIONS (4X10=40)	CO	BL	Marks
✓ 1	a) Define rubbing velocity at a pin joint. What will be the rubbing velocity at pin joint when the two links move in the same and opposite directions ? b) State Aronhold Kennedy theorem	CO2	L2	5
2	Find the linear velocities at points A,B,C,and D and angular velocities of the links AB,BC,and CD for the above linkage using ICR method if crank OA rotates clockwise at 150rpm 	CO1 CO2	L1 L3	5 10
3	Find the linear velocity of the slider F and angular velocities of the links AB,BC,CD and EF using relative velocity method if crank OA rotates clockwise at 120rpm AB=DE=150mm, BC=CD=450mm, EF=375mm 	CO2	L3	10

- 4 The dimensions and configuration of the four bar mechanism, shown in Fig. are as follows :
- $P_1A = 300 \text{ mm}$; $P_2B = 360 \text{ mm}$; $AB = 360 \text{ mm}$, and $P_1P_2 = 600 \text{ mm}$.
 The angle $AP_1P_2 = 60^\circ$. The crank P_1A has an angular velocity of 10 rad/s and an angular acceleration of 30 rad/s^2 , both clockwise.
 Determine the angular velocities and angular accelerations of P_2B , and AB and the velocity and acceleration of the joint B .



CO2

L2

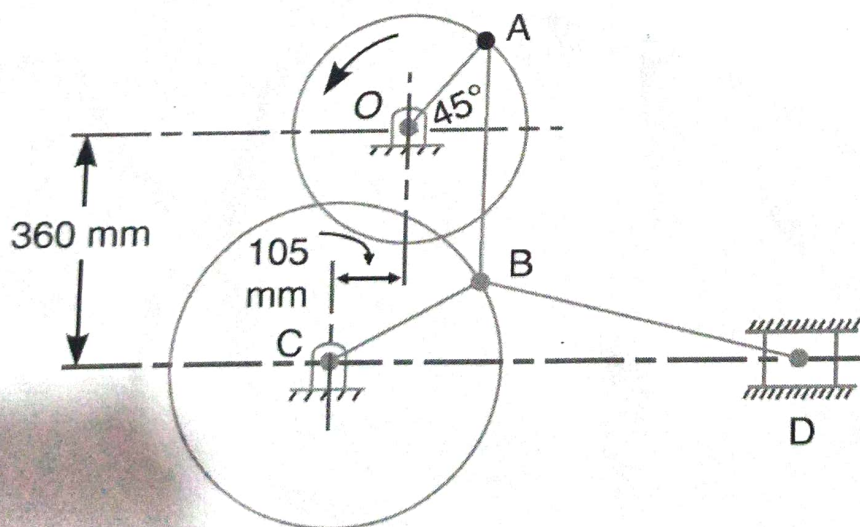
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- 5 In the following toggle mechanism, the slider D is constrained to move on a horizontal path. The crank OA is rotating in the counter-clockwise direction at a speed of 180 r.p.m. increasing at the rate of 50 rad/s^2 . The dimensions of the various links are as follows:

$OA = 180 \text{ mm}$; $CB = 240 \text{ mm}$; $AB = 360 \text{ mm}$; and $BD = 540 \text{ mm}$.

For the given configuration, find

1. Velocity of slider D and angular velocity of BD , and
2. Acceleration of slider D and angular acceleration of BD .



CO2

L3

10

Fifth Semester B.Tech. in Mechanical Engineering

First Internal Examination

1503 Machine Tools and Machining Science

Time: 2 hrs

Max: 50 marks

Answer all questions

- ✓ 1. (a) Explain with the help of sketches what you understand by the term orthogonal cutting and its relevance in metal cutting theory. (5)
(b) What are the conditions that would allow a continuous chip to be formed (3)
(c) From the machining performance viewpoint which type of chip is preferred (2)
- ✓ 2. (a) Show the sketches of a single point cutting tool (SPCT) showing its shank and other important faces. (3)
(b) Draw the tool geometry of a SPCT. Show the important tool angles. (4)
(c) Explain the significance of important tool angles in SPCT and how they affect the cutting performance. (3)
- ✓ 3. (a) Show schematically the Merchant's force circle in orthogonal cutting. (3)
(b) Derive the equations for shear and friction forces using Merchant's circle (3)
(c) In an orthogonal cutting test with a tool of rake angle 10° , the following observations were made: chip thickness ratio = 0.3, cutting force (horizontal) = 1290 N, cutting force (vertical) = 1650 N. Calculate the components of cutting forces and the coefficient of friction at the chip-tool interface. (4)
- ✓ 4. (a) During orthogonal machining operation on mild steel, the following results were obtained: $t_1 = 0.25$ mm, $t_2 = 0.75$ mm, $w = 2.5$ mm, rake angle = 5° , $F_c = 950$ N, $F_t = 475$ N. Determine the ultimate shear stress of the work material. (7)
(c) Discuss the nature of friction found in metal cutting. (3)
5. (a) Explain how the chip is formed with the help of suitable sketches. (3)
(b) Explain how the tool angles of a SPCT are specified in ASA system. (4)
(c) What do you understand by the term *tool signature*? (3)

School of Engineering CUSAT

First internal Examination-Oct 2024

19-205-0504 Thermal Engineering

Time: 2 Hrs (Answer all Questions Max marks 30)

	PART A	Mark	BL	CO	PO
✓ 1.	Explain enthalpy of formation and enthalpy of combustion.	3	1	1	1
✓ 2.	Derive the air standard efficiency of Diesel cycle in terms of r and r_c	3	2	1	1
✓ 3.	Explain the Ignition system of a 4 stroke 6 cylinder SI engine	3	1	2	1
4	Explain Wankel Engine and Stratified charged engine.	3	1	3	1

	PART B	Mark	BL	CO	PO
✓ 5	Octane (C_8H_{18}) is burned with dry air. The volumetric analysis of the products on a dry basis is 9.21 percent CO_2 , 0.61 percent CO , 7.06 percent O_2 , and 83.12 percent N_2 . Determine (a) the air-fuel ratio and (b) the percentage of theoretical air used.	12	2	2	1
✓ 6	A fourstroke dual combustion cycle operates on 1 kg of air per second at 1 bar and 27deg C per cycle. The addition of heat at constant volume is 24000 kJ /kg and constant volume pressure ratio due to heat addition ie $r_p=2$. And $r=15$. Determine the statepoints at all points in the cycle, cut off ratio, thermal efficiency and mean effective pressure.	12	2	2	1

ODD NOS
School of Engineering
Cochin University of Science And Technology
B.Tech Degree Vth Semester first series test in Mechanical Engineering (A&B)
 Time: 2Hrs Sub: **19-205-0505:POWER PLANT ENGINEERING** Max.Marks:25
 (Answer all questions , *Answer Section A and B in separate answer books*)

Q. No	Section A
1	Define the terms Demand factor, Plant use factor & Diversity factor. Explain each with an example.(3 Marks)
(a)	
(b)	Fill in the blanks (All question carries one mark) (i) The connected load is generally than the maximum demand. (ii) The value of demand factor is than 1. (iii) The higher the load factor of a power station, the is the cost per unit generated. (iv) The value of diversity factor is than 1. (2 Marks)
(c)	Explain with diagram ,the function of a Draft Tube . (4 Marks)
2	Air enters the compressor of an air standard Brayton cycle at 0.2 Mpa and 12°C. The pressure leaving the compressor is 1.3 MPa, and the maximum temperature in the cycle is 1150°C. If a re-generator with effectiveness 0.9 is incorporated into the cycle, determine (i) The pressure and temperature at each point in the cycle. (ii) The compressor work, turbine work, and cycle efficiency. Assume the isentropic compressor efficiency of 75% and isentropic turbine efficiency of 82% . (8 Marks)
(a)	
(b)	Explain the advantages of the process of reheating in Brayton cycle using p-v and T-s diagram. (3 Marks)
(c)	What are the major components in a Diesel Engine Power Plant ? What are the different starting methods used in Diesel Engine Power Plants ?(5Marks)

Section B

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING
B. TECH DEGREE IN MECHANICAL ENGINEERING
V-SEMESTER FIRST INTERNAL EXAMINATIONS, OCTOBER 2024
19-205-0605 (IE): INDUSTRIAL MANAGEMENT

TIME: 2 Hrs

MAX. MARKS: 40

Course Outcomes

On completion of this course the student will be able to:

CO1	Understand the concept of management, organization and personal management and apply those concepts in industrial environment
CO2	Get the concept of industrial relations and use it in real life
CO3	Understand the concept of financial management, project management and economics to use in an industrial situation.
CO4	Study the concept of marketing management and basics of IPR.

PART A

(Answer all questions)

(4 X 4 marks= 16 Marks)

Q.No	Questions	Marks	BL	CO	PO
I (a)	Explain system approach of management	4	L1	CO1	PO1,PO2
(b)	Explain the contributions of Gilberth and Gantt	4	L1	CO1	PO1,PO2
(c)	Draw and explain supply, demand and equilibrium curves.	4	L2	CO2	PO1,PO2
(d)	Explain the purpose and format of journals and ledgers in financial accounting.	4	L2	CO2	PO1,PO2

PART B

(Answer the following questions)

(2 x 12 marks= 24 marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	i) Define organization structure. ii) What are the commonly followed organizational structures in industry? iii) Mention its advantages, disadvantages and application.	12	L1	CO1	PO1, PO2
OR					
III (a)	Discuss the skills required for an entry- level position with respect to the priority order.	6	L2	CO1	PO1, PO2